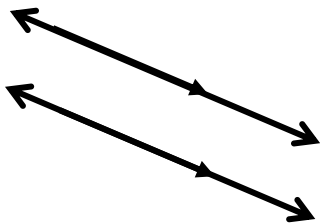
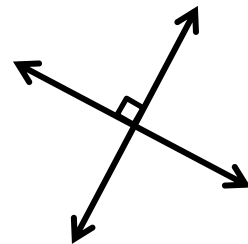


Slopes of Parallel Lines



Slopes of Perpendicular Lines

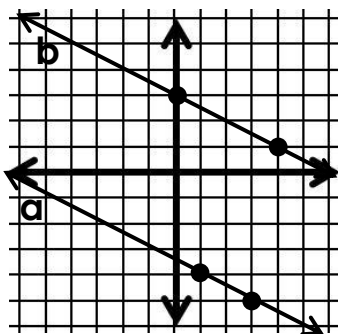


Slope of Parallel Lines

In a coordinate plane, two non-vertical lines are parallel if and only if they have the same slope. Any two vertical lines are parallel.

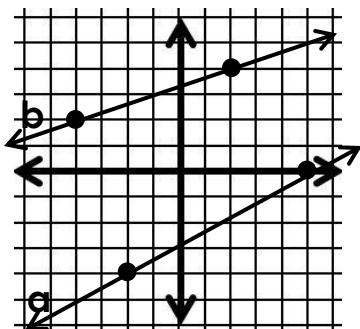
Example 1:

Find the slope of each line. Is $a \parallel b$?



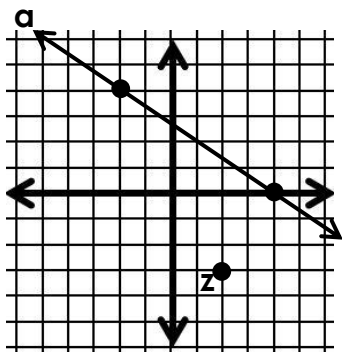
Example 2:

Find the slope of each line. Is $a \parallel b$?



Example 3:

Draw a line that is parallel to line a and passes through point z.

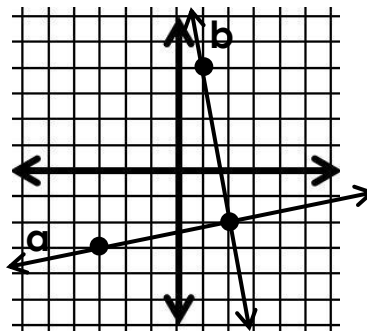


Slope of Perpendicular Lines

In a coordinate plane, two non-vertical lines are perpendicular if and only if the product of their slopes is -1. Horizontal lines are perpendicular to vertical lines.

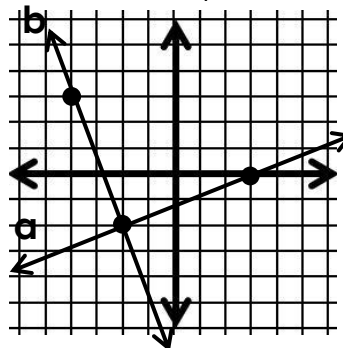
Example 4:

Find the slope of each line. Is $a \perp b$?



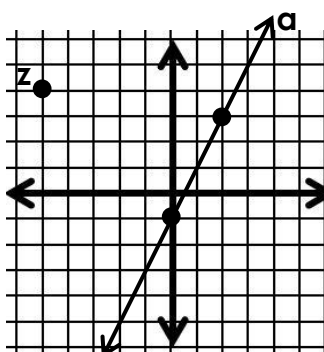
Example 5:

Find the slope of each line. Is $a \perp b$?



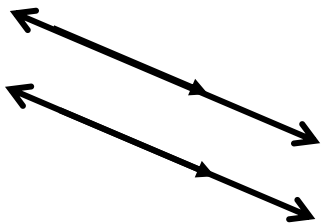
Example 6:

Draw a line that is perpendicular to line a and passes through point z.

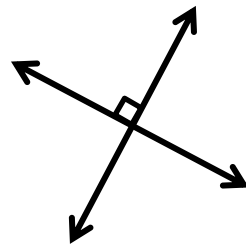


Recall: $\text{Slope} = m = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$

Slopes of Parallel Lines



Slopes of Perpendicular Lines



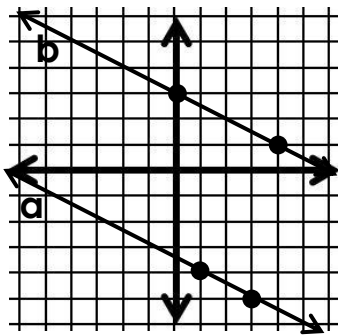
Slope of Parallel Lines

In a coordinate plane, two non-vertical lines are parallel if and only if:

Any two vertical lines are parallel.

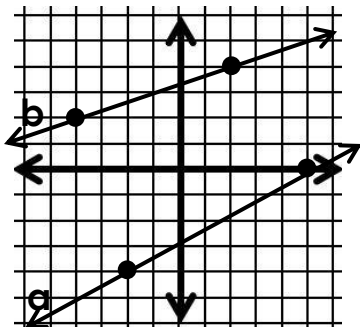
Example 1:

Find the slope of each line. Is $a \parallel b$?



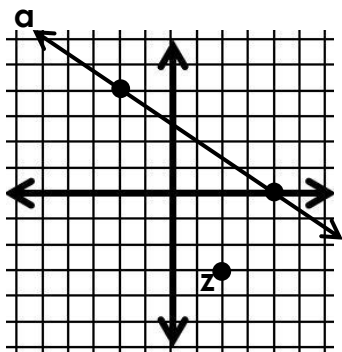
Example 2:

Find the slope of each line. Is $a \parallel b$?



Example 3:

Draw a line that is parallel to line a and passes through point z.



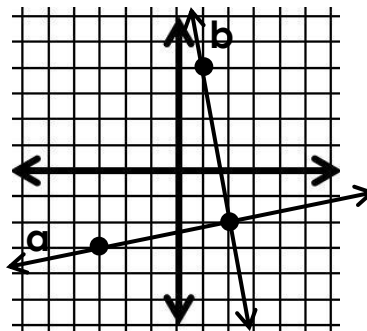
Slope of Perpendicular Lines

In a coordinate plane, two non-vertical lines are perpendicular if and only if:

Horizontal lines are perpendicular to vertical lines.

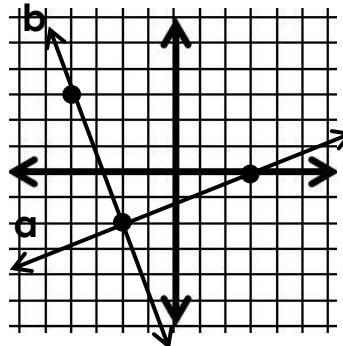
Example 4:

Find the slope of each line. Is $a \perp b$?



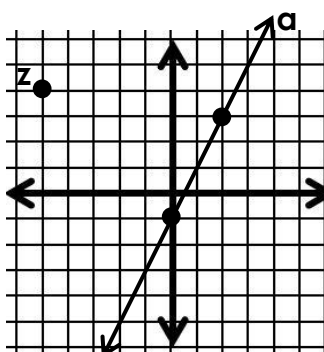
Example 5:

Find the slope of each line. Is $a \perp b$?



Example 6:

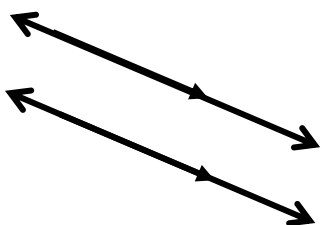
Draw a line that is perpendicular to line a and passes through point z.



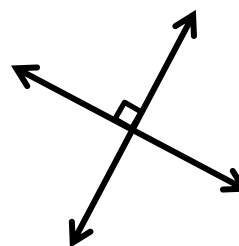
Recall:
$$\text{Slope} = m = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Answer Key!

Slopes of
Parallel
Lines



Slopes of
Perpendicular
Lines

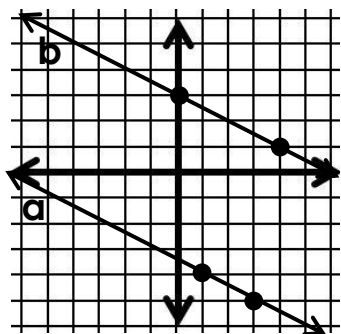


Slope of Parallel Lines

In a coordinate plane, two non-vertical lines are parallel if and only if they have the same slope. Any two vertical lines are parallel.

Example 1:

Find the slope of each line. Is $a \parallel b$?



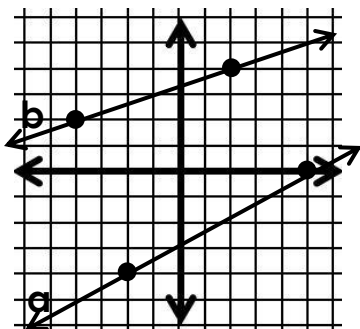
line a:
 $m = \frac{-1}{2}$

line b:
 $m = \frac{-1}{2}$

YES, $a \parallel b$

Example 2:

Find the slope of each line. Is $a \parallel b$?



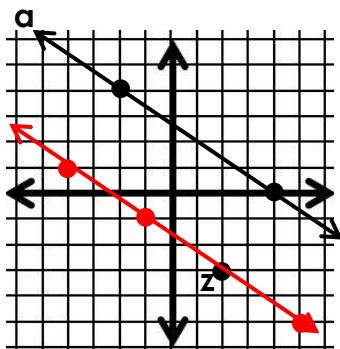
line a:
 $m = \frac{4}{7}$

line b:
 $m = \frac{1}{3}$

NO, $a \not\parallel b$

Example 3:

Draw a line that is parallel to line a and passes through point z.



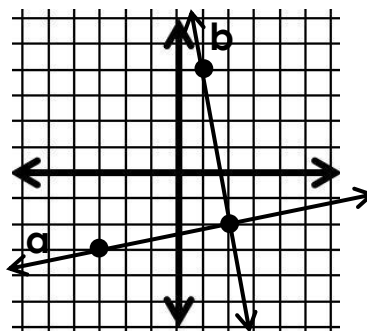
line a:
 $m = \frac{-2}{3}$

Slope of Perpendicular Lines

In a coordinate plane, two non-vertical lines are perpendicular if and only if the product of their slopes is -1. Horizontal lines are perpendicular to vertical lines.

Example 4:

Find the slope of each line. Is $a \perp b$?



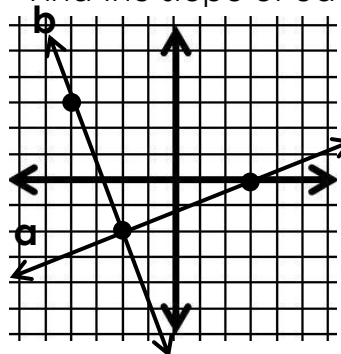
line a:
 $m = \frac{1}{2}$

line b:
 $m = -\frac{1}{2}$

No, not perpendicular

Example 5:

Find the slope of each line. Is $a \perp b$?



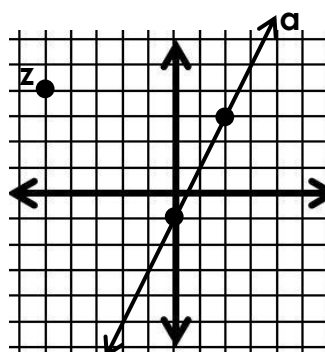
line a:
 $m = \frac{2}{5}$

line b:
 $m = \frac{-5}{2}$

YES, $a \perp b$

Example 6:

Draw a line that is perpendicular to line a and passes through point z.



line a:
 $m = 2$

new line
 $m = \frac{-1}{2}$

Recall: $\text{Slope} = m = \frac{\text{change in } y}{\text{change in } x} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$

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Directions

Photo copy pages 1 & 2 front to back (3 & 4 if you want students to write in their own notes at the top; 5 & 6 for the answer key). On my printer, I use the option: double sided and to flip along the short edge.

Have students cut along the dotted lines to create two tabs.

The final product should look like this:

